Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A number x is chosen at random from the set $1, 2, 3, 4, \ldots \ldots, 100$. Define the event: A = the chosen number x satisfies $\frac{x - 10x - 50}{x - 30} \geq 0$, then PA is

(A) 0.20

(B) 0.51

(C) 0.71

(D) 0.70

Q.2

12 people are asked questions in succession in a random order and exactly 3 out of 12 people know the answer. The probability that the 6th person asked is the 2nd person to know the answer, is

(A) $\frac{10}{21}$

(B) $\frac{3}{22}$

(C) $\frac{7}{11}$

(D) $\frac{5}{12}$

Q.3

If A, B, C are three events such that $PB = \frac{4}{5}$, $PA \cap B \cap C^c = \frac{1}{4}$ and $PA^c \cap B \cap C^c = \frac{1}{3}$, then $PB \cap C$ is equal to

(A) $\frac{11}{60}$

(B) $\frac{1}{5}$

(C) $\frac{13}{60}$

(D) $\frac{1}{4}$

Q.4

Amar, Bimal and Chetan are three contestants for an election, odds against Amar will win is $4:1$ and odds against Bimal will win is $5:1$ and odds in favor of Chetan will win $2:3$ then what is probability that either Amar or Bimal or Chetan will win the election.

(A) $23/20$

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(B) $11/30$

(C) $7/10$

(D) None of these

Q.5

A coin is tossed three times. Consider the following events:
A: No head appears
B: Exactly one head appears
C: At least two heads appear Which one of the following is correct?

(A) $(\mathrm{A} \cup \mathrm{B}) \cap (\mathrm{A} \cup \mathrm{C}) = \mathrm{B} \cup \mathrm{C}$

(B) $(\mathrm{A} \cap \mathrm{B}') \cup (\mathrm{A} \cap \mathrm{C}') = \mathrm{B}' \cup \mathrm{C}'$

(C) $\mathrm{A} \cap (\mathrm{B}' \cup \mathrm{C}') = \mathrm{A} \cup \mathrm{B} \cup \mathrm{C}$

(D) $\mathrm{A} \cap (\mathrm{B}' \cup \mathrm{C}') = \mathrm{B}' \cap \mathrm{C}'$

Q.6

Let A and B be two events such that
$P(A \cap B') = 0.20, P(A' \cap B) = 0.15$
$P(A' \cap B') = 0.1$, then $P(A/B)$ is equal to

(A) $11/14$

(B) $2/11$

(C) $2/7$

(D) $1/7$

Q.7

Two aeroplanes I and II bomb a target in succession. The probabilities of I and II scoring a hit correctly are 0.3 and 0.2, respectively. The second plane will bomb only if the first misses the target. The probability that the target is hit by the second plane is

(A) 0.2

(B) 0.7

(C) 0.06

(D) 0.14

Q.8

Let $0 < \mathrm{P(A)} < 1, 0 < \mathrm{P(B)} < 1$ and
$\mathrm{P(A \cup B)} = \mathrm{P(A)} + \mathrm{P(B)} - \mathrm{P(A)P(B)}$, then

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(A) $P(B/A) = P(B) - P(A)$

(B) $P(A' \cup B') = P(A') + P(B')$

(C) $P(A \cap B) = P(A')\,P(B')$

(D) None of these

Q.9

n letters to each of which corresponds on addressed envelope are placed in the envelope at random. Then the probability that n^{th} letter is placed in the right envelope, will be :

(A) $\frac{1}{1!} - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \ldots . (-1)^n \frac{1}{n!}$

(B) $\frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \ldots \cdot \frac{1}{n!}$

(C) $\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \ldots . (-1)^n \frac{1}{n!}$

(D) None of these

Q.10

Two dice are thrown n times in succession. The probability of obtaining a double-six atleast once is

(A) $\left(\frac{1}{36}\right)^n$

(B) $1 - \left(\frac{35}{36}\right)^n$

(C) $\left(\frac{1}{12}\right)^n$

(D) None of these

Q.11

A bag contains p white and q black ball. Two players A and B alternately draw a ball from the bag, replacing the balls each time after the draw till one of them draws a white ball and wins the game. If A begins the game and the probability of A winning the game is three times that of B,
then the ratio $p : q$ is :

(A) $3 : 4$

(B) $4 : 3$

(C) $2 : 1$

(D) $1 : 2$

Q.12

A box contains 20 identical balls of which 10 are blue and 10 are green. The balls are drawn at random from the box one at a time with replacement. The probability that a blue ball is drawn 4 th time on the 7 th draw is

(A) $\frac{27}{32}$

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(B) $\frac{5}{64}$

(C) $\frac{5}{32}$

(D) $\frac{1}{2}$

Q.13

Three houses are available in a locality. Three persons apply for the houses. Each applies for one house without consulting others. The probability that all the three apply for the same house is

(A) $\frac{8}{9}$

(B) $\frac{7}{9}$

(C) $\frac{2}{9}$

(D) $\frac{1}{9}$

Q.14

Five horses are in a race. Mr. A selects two of the horses at random and bets on them. The probability that Mr. A selected the winning horse is

(A) $\frac{4}{5}$

(B) $\frac{3}{5}$

(C) $\frac{1}{5}$

(D) $\frac{2}{5}$

Q.15

Probability that India will win against Pakistan in a cricket match is $2/3$, in series of 5 matches what is the probability that India will win the series?

(A) $161/81$

(B) $192/243$

(C) $172/243$

(D) None of these

Q.16

An urn contains nine balls of which three are red, four are blue and two are green. Three balls are drawn at random without replacement from the urn. The probability that the three balls have different colour is

(A) $2/7$

(B) $1/21$

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(C) $2/23$

(D) $1/3$

Q.17

If $x = 33^n, n$ is a positive integral value, then what is the probability that x will have 3 at its units place, when n is selected at random

(A) $1/3$

(B) $1/4$

(C) $1/5$

(D) $1/2$

Q.18

A number is chosen randomly from the set $\{1, 2, 3, \ldots, 15\}$. If chosen number is a two digit number, then the probability that the number is a prime numbers, is

(A) $1/2$

(B) $1/3$

(C) $1/4$

(D) $1/5$

Q.19

If the integers m and n are chosen at random from 1 and 100 , then the probability that a number of from $7^{\mathrm{m}} + 7^{\mathrm{n}}$ is divisible by 5 , equals

(A) $1/4$

(B) $1/7$

(C) $1/8$

(D) $1/49$

Q.20

The probability that two persons have their same birthday is (considering year to be of 365 days)

(A) $\frac{364}{365}$

(B) $\left(\frac{1}{365}\right)^2$

(C) $\frac{1}{365}$

(D) $\left(\frac{1}{365}\right)^3$

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Q.1 Answer:

0.71

Solution:

$S = 1, 2, 3 \ldots \ldots 100$

A: **Chosen no.** x **satisfies**

$\frac{x - 10x - 50}{x - 30} \geq 0$

$\therefore x \in 10, 11, 12 \ldots \ldots 29 \cup 50, 51 \ldots \ldots 100$

$PA = \frac{71}{100} = 0.71$

Q.2 Answer:

$\frac{3}{22}$

Solution:

6th **Person asked is the** 2nd **person to know the answer**

⇒ **In the first five people there must be exactly one person who knows the answer**

Required probability $= \frac{{}^{9}C_4 \times {}^{3}C_1}{{}^{12}C_5} \times \frac{2}{7}$

$= \frac{9 \times 8 \times 7 \times 6 \times 120 \times 3}{24 \times 12 \times 11 \times 10 \times 9 \times 8} \times \frac{2}{7} = \frac{3}{22}$

Q.3 Answer:

$\frac{13}{60}$

Solution:

From set theory

$PB = PA \cap B \cap C^c + A^c \cap B \cap C^c + PB \cap C$

$\frac{4}{5} = \frac{1}{4} + \frac{1}{3} + PB \cap C$

$\Rightarrow PB \cap C = \frac{4}{5} - \frac{1}{4} - \frac{1}{3} = \frac{48 - 15 - 20}{60}$

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$= \frac{13}{60}$

Q.4 Answer:

None of these

Solution:

From the given information probability that Amar will win the tournament is $\mathrm{P}(\mathrm{A}) = 1/5$ and Bimal will win is $\mathrm{P}(\mathrm{B}) = 1/6$ and same for Chetan is $\mathrm{P}(\mathrm{C}) = 2/5$
Since these events are mutually exclusive.

Q.5 Answer:

$\mathrm{A} \cap (\mathrm{B}' \cup \mathrm{C}') = \mathrm{B}' \cap \mathrm{C}'$

Solution:

$U = \{(HHH)(HHT)(HTH)(HTT)$
$(THH)(THT)(TTH)(TTT)\}$
$A = \{(TTT)\}$
$B = \{(HTT)(THT)(TTH)\}$
$C = \{(HHH)(HHT)(HTH)(THH)\}$
By checking the options
(d) $A \cap (B' \cup C') = B' \cap C'$ is correct.

Q.6 Answer:

$11/14$

Solution:

$P(A \cap B') = P(A) - P(A \cap B) = 0.20$
Also, $P(A' \cap B) = P(B) - P(A \cap B) = 0.15$
$\Rightarrow P(A) + P(B) - 2P(A - B) = 0.35$
Now, $P(A' \cap B') = 1 - P(A \cup B)$
$\Rightarrow 0.1 = 1 - P(A) - P(B) + P(A \cap B)$
$\Rightarrow P(A) + P(B) - P(A \cap B) = 0.9$
$\Rightarrow P(A \cap B) = 0.9 - 0.35 = 0.55$ **and**
$P(A) = 0.75, P(B) = 0.70$
Now, $P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{0.55}{0.70}$.

Q.7 Answer:

0.14

Solution:

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Given : Probability of aeroplane I, scoring a target correctly i.e., $P(I) = 0.3$ probability of scoring a target correctly by aeroplane II, i.e. $P(II) = 0.2$

$\therefore P(\bar{I}) = 1 - 0.3 = 0.7$

$\therefore$ **The required probability** $= \mathrm{P}(\bar{I} \cap II) = P(\bar{I}) \cdot P(II) = 0.7 \times 0.2 = 0.14$

Q.8 Answer:

None of these

Solution:

Given $\mathrm{P(A) + P(B) - P(A)P(B) = P(A \cup B)}$

Comparing with $\mathrm{P(A) + P(B) - P(A \cap B) = P(A \cup B)}$

we get $\mathrm{P(A \cap B) = P(A) \cdot P(B)}$

$\therefore$ A **and** B **independent events.**

Q.9 Answer:

$\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \ldots . (-1)^{\mathrm{n}} \frac{1}{\mathrm{n!}}$

Solution:

Probability of n letters $= 1 - \mathrm{P}\left(\mathrm{EA_1 \dot{E} A_2 E A_3 \ldots \ EA_n}\right.$

Where $A_1 \ldots . . A_n$ **the event that the letter is placed at right envelope.** $= 1 -$

$[\Sigma \mathrm{P}\left(\mathrm{A_i}\right) - \Sigma \mathrm{P}\left(\mathrm{A_i} \cap \mathrm{A_k}\right)$

$+ \sum \mathrm{P}\left(\mathrm{A_i} \cap \mathrm{A_j} \cap \mathrm{A_k}\right) \ldots \ldots + (-1)^{\mathrm{n}-1} \mathrm{P}\left(\mathrm{A_i} \cap \mathrm{A_j} \cap \mathrm{A_n}\right)]$

Here, $\mathrm{P}\left(\mathrm{A_i}\right) = \frac{(\mathrm{n}-1)!}{\mathrm{n!}}$

$P\left(A_1 \cap A_2 \cap A_3 \cap \ldots \ldots \cap A_n\right) = \frac{(n-r)!}{n!}$

$\Rightarrow \quad \Sigma \overline{\mathrm{A}}_1 \cap \overline{\mathrm{A}}_2 \cap \overline{\mathrm{A}}_3 \cap \ldots \ldots \ldots \cap \overline{\mathrm{A}}_{\mathrm{n}}$

$= 1 - \left[\frac{1}{1!} - \frac{1}{2!} + \frac{1}{3!} \ldots . . (-1) \frac{\mathrm{n}-1!}{\mathrm{n!}}\right]$

$= \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \ldots \ldots \ldots + (-1)^{\mathrm{n}} \frac{1}{\mathrm{n!}}$

Q.10 Answer:

$1 - \left(\frac{35}{36}\right)^n$

Solution:

The probability of getting a double -six in one throw $= \frac{1}{36}$

The probability of not getting a double -six in one throw $= 1 - \frac{1}{36} = \frac{35}{36}$

So the probability of not getting a double -six in

n **throw** $= \left(\frac{35}{36}\right)^n$

$\therefore$ **Probability of obtaining a double-six atleast once** $= 1 - \left(\frac{35}{36}\right)^n$

Q.11 Answer:

$2 : 1$

Solution:

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The probability that A wins is:

$(a/a+b)+(b/a+b)*(b/a+b)*(a/a+b)+\ldots\ldots\ldots\ldots$ **ill infinity**

taking out $a/a+b$ **common and then applying the formula of infinite gp series**

we have $P(A)=a+b/a+2\,b$

Now, finding the probability of B

$(b/a+b)*(a/a+b)+(b/a+b)*(b/a+b)*(b/a+b)*(a/a+b)+$

$\ldots\ldots\ldots\ldots\ldots$ **till infinity**

Taking out $(b/a+b)*(a/a+b)$ **common and applying the formula of infinite gp**

we have $P(B)=\frac{b}{a}+2\,b$

Now in the question it has been clearly mentioned that $P(A)=3P(B)$

...therefore by solving it we are getting the ratio to be $2:1$

Q.12 Answer:

$\frac{5}{32}$

Solution:

Probability of getting a blue ball at any draw $=p=\frac{10}{20}=\frac{1}{2}$

[getting a blue ball 4 th time in 7 th draw $]=P$ **[getting 3 blue balls in 6 draws** $]\times P$ **[a blue ball in the 7 th draw].** $={}^{6}C_{3}\left(\frac{1}{2}\right)^{3}\left(\frac{1}{2}\right)^{3}\cdot\frac{1}{2}$

$=\frac{6\times5\times4}{1\times2\times3}\left(\frac{1}{2}\right)^{7}=20\times\frac{1}{32\times4}=\frac{5}{32}$

Q.13 Answer:

$\frac{1}{9}$

Solution:

Number of points in the sample space is

27 and the number of points favourable to the event is 3 . Hence, the required probability is

$\frac{1}{9}$

Q.14 Answer:

$\frac{2}{5}$

Solution:

Select 2 out of $5=\frac{2}{5}$

Q.15 Answer:

$192/243$

Solution:

Total number of matches $=n=5$ **India will win the series if it wins either 3 or 4 or matches. In previous question we have calculated the value of** $P(3)=$ **probability of winning 3 matches** $=$

$\left({}^{5}C_{3}\right)\left(\frac{2}{3}\right)^{3}\left(\frac{1}{3}\right)^{2}$

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Required Probability $= \mathrm{P}(3)+\mathrm{P}(4)+\mathrm{P}(5)$
$= \left({}^{5}\mathrm{C}_{3}\right)\left(\frac{2}{3}\right)^{3}\left(\frac{1}{3}\right)^{2}+\left({}^{5}\mathrm{C}_{4}\right)\left(\frac{2}{3}\right)^{4}\left(\frac{1}{3}\right)^{1}$
$+\left({}^{5}\mathrm{C}_{5}\right)\left(\frac{2}{3}\right)^{5}\left(\frac{1}{3}\right)^{0}$
$=\frac{10'8}{243}+\frac{5'16}{243}+\frac{1'32}{243}=\frac{192}{243}$

Q.16 Answer:

$2/7$

Solution:

$\mathbf{n(S)} = {}^{9}\mathbf{C}_{3}$
$\mathbf{n(E)} = {}^{3}\mathbf{C_T} \times {}^{4}\mathbf{C_1} \times {}^{2}\mathbf{C}$
Probability $== \frac{3\times4\times2}{{}^{9}\mathrm{C}_{3}} = \frac{24\times3!}{9!} \times 6!$
$= \frac{24\times6}{9\times8\times7} = \frac{2}{7}$

Q.17 Answer:

$1/4$

Solution:

$x = 33^{n}$
Only four digits may be at the units. Place which are $1, 3, 7, 9$, **when** n **is a positive integral value.**
$\therefore$ **Probability getting 3 at unit's place is** $1/4$

Q.18 Answer:

$1/3$

Solution:

Required probability $= 2/6 = 1/3$

Q.19 Answer:

$1/4$

Solution:

Total number of possible ways
$= 100 \times 100$
For $7^{\mathrm{m}}+7^{\mathrm{n}}$ **to be divided by 5, (i) one of the term has to end with 9 and other with**
1 **m can be** $2, 6, 10, 14, \ldots, 98(25$ **values** $)$ **n can be** $4, 8, 12, \ldots, 100$ **(25 values)**
(ii) one of the term has to end with 3 and other with 3
Probability $= \frac{2\times2\times25\times25}{100\times100} = \frac{1}{4}$

Q.20 Answer:

$\frac{1}{365}$

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Solution:

First person is born on any day but second has same birthday its chance is $\frac{1}{365}$.

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